

CLINICAL PRACTICE

Prevention and Treatment of Peanut Allergy

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This Journal feature begins with a case vignette highlighting a common clinical problem. Evidence supporting various strategies is then presented, followed by a review of formal guidelines, when they exist. The article ends with the authors' clinical recommendations.

SUMMARY

Early introduction of peanut protein reduces allergy prevalence by approximately 80%, with efficacy diminishing as introduction is delayed. Appropriate prevention involves ingestion of approximately 2 g of peanut protein weekly for infants at low risk and 4 to 6 g weekly for infants at high risk. Population-level implementation that targets all infants achieves greater reduction in disease burden than approaches that target only high-risk groups, although disparities exist among some ethnic groups and groups with restricted access to care. Peanut immunotherapy initiated in younger children (1 to 3 years of age) shows superior efficacy and higher rates of clinical remission as compared with immunotherapy initiated in older children. The natural history of untreated peanut allergy follows a trajectory of increasing peanut-specific IgE levels and clinical reactivity over time, underscoring the importance of early intervention during this narrow developmental window.

A 4-month-old infant of East Asian heritage had been brought to the clinic for routine care. The infant had severe, generalized atopic dermatitis that developed at 6 weeks of age and had been treated with topical glucocorticoid agents. The infant was exclusively breast-fed, and there was a plan to continue breast-feeding until the child reached 6 months of age. The parents asked about introducing solid foods and preventing food allergies. The pediatrician recommended including smooth peanut butter in the infant's diet when weaning to prevent peanut allergy. Now, 3 months later, the family returns and reports that hives and vomiting had developed after the first introduction of peanut butter at 7 months of age. Allergy tests are performed, and a diagnosis of peanut allergy is confirmed by a positive skin-prick test (wheal diameter, 6 mm) and peanut-specific IgE level of 1.8 kU per liter. How would you counsel parents about reducing the risk of peanut allergy, and how would you manage this child's peanut allergy?

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THE CLINICAL PROBLEM

PEANUT ALLERGY IS AN IGE-MEDIATED ALLERGIC CONDITION AND IS ONE of the most common food allergies, with prevalence increasing substantially over the past four decades.^{1,2} Peanut allergy typically manifests in early childhood and affects approximately 2% of children in high-income countries. Unlike other common food allergies, such as egg and milk allergies, peanut allergy has greater persistence, with approximately only 20 to 32% of cases resolving naturally by the time the child is 6 to 10 years of age.^{3,4}

A diagnosis of peanut allergy is made by identifying a history of allergic symptoms that usually occur within 2 hours after peanut ingestion and confirming IgE sensitization with either a positive peanut skin-prick test or detection of peanut-specific IgE and peanut components, such as Ara h 2 antibodies in serum. Oral food challenge is reserved for cases in which clinical history and testing are inconclusive.

Food allergy typically develops during the first year of life. More specifically, 65 to 70% of cases of peanut allergy develop by 1 year of age.⁴⁻⁶ Allergic reactions in infants younger than 1 year of age are generally mild,^{7,8} and many children 1 to 4 years of age with peanut allergy only have reactions to high doses of peanut.⁹ Exposure to peanut protein in persons with allergy can trigger life-threatening reactions. Given the common inclusion of peanuts in foods, persons allergic to peanut protein must have constant access to epinephrine for acute treatment of reactions caused by accidental exposure. Generally, in persons with untreated peanut allergy, sensitivity to small doses of peanut and the severity of reactions increase with age, and the majority of fatalities, albeit rare, occur in young adulthood.¹⁰

In children with peanut allergy, the increase in clinical reactivity to peanut that comes with age is mirrored by a progressive increase in Ara h 2–specific IgE levels and expansion in the size of wheals from a peanut skin-prick test, which persist despite strict dietary avoidance (Fig. 1).⁷ This sustained IgE response is in contrast to typical humoral immune responses and other hypersensitivity states. In the case of routine childhood immunizations, periodic booster injections are needed to prevent IgG levels and clinical protection from waning. In celiac disease — a small-bowel hypersensitivity reaction to dietary gluten — tissue transglutaminase IgA typically becomes undetectable within a year after the initiation of a gluten-free diet. However, in patients with peanut allergy, Ara h 2–specific IgE synthesis continues to escalate in the absence of ongoing peanut exposure. Furthermore, the nature of the IgE response to peanut changes over time, with IgE initially being directed at conformational epitopes until approximately 2.5 years of age, after which IgE to linear epitopes (ones that are associated with persistent food allergy) develops.¹² This progressive response raises the question of whether earlier treatment

in infants who have peanut allergy — before they reach 2.5 years of age — could be potentially curative.

Risk factors for peanut allergy include genetic predispositions, ethnic group, male sex, early-onset atopic dermatitis, skin *Staphylococcus aureus* colonization, dietary factors, and changes in environmental exposures (Fig. 2).^{3,6,13-16} The dual-allergen–exposure hypothesis postulates that oral exposure to an allergen promotes antigen tolerance, whereas isolated cutaneous exposure to an allergen (e.g., through a compromised skin barrier) in the absence of oral exposure promotes sensitization and subsequent development of allergy.¹⁷ In one prospective cohort study, infants with severe atopic dermatitis who were treated with emollients containing peanut oil in the first 6 months of life were at a significantly increased risk for peanut allergy.¹⁸

The overall hypothesis is best illustrated in a study by Fox et al.,¹⁹ which showed that increasing environmental exposure to peanut in children with atopic conditions was associated with an increased risk of peanut allergy among children who did not eat peanuts. However, early introduction of peanut in the first year of life attenuated this relationship and decreased the risk of peanut allergy.¹⁹ Randomized clinical trials that tested induction of oral tolerance (an active antigen-specific state in which a food can be consumed without allergic symptoms resulting) to peanut^{7,20,21} showed substantial prevention of peanut allergy and have led to a shift in allergy prevention guidelines, with international guidelines advocating for early peanut introduction in infants who are weaning to prevent the development of peanut allergy.^{22,23}

STRATEGIES AND EVIDENCE

PRIMARY AND SECONDARY PREVENTION OF PEANUT ALLERGY

Effective prevention strategies are a public health priority.⁵ Prevention of peanut allergy can be categorized as primary (preventing initial peanut sensitization) or secondary (preventing peanut allergy in sensitized children); the tertiary strategy involves reducing the severity and frequency of reactions to peanut.

Clinical trials have shown efficacy of both primary and secondary prevention strategies for peanut allergy (Table 1). Early introduction of

6 g of peanut protein per week in young infants with atopic dermatitis or egg allergy (or both) reduced the incidence of peanut allergy by 81% in the Learning Early About Peanut Allergy (LEAP) trial,⁷ and in the Enquiring about Tolerance (EAT) trial, early introduction of 4 g of peanut protein per week in breast-fed infants 3 to 6 months of age was also effective in preventing peanut allergy.²⁰ In the LEAP trial, exposure to peanut protein was effective in both primary and secondary prevention. The children included in the LEAP trial remained protected from peanut allergy after avoiding peanut for 1 year (from 5 to 6 years of age),²⁴ with evidence of protection enduring into adolescence after a 6-year period of ad libitum peanut exposure.²⁵

Integrated data from the EAT and LEAP trials confirm that even within an infant population at lower risk, early allergen introduction remains protective across all risk subgroups, including among infants with no atopic dermatitis and of different races.²⁶ Further analysis points to increased intervention efficacy with earlier age of introduction, especially in infants with atopic dermatitis.⁶ Additional trials, including the Prevention of Allergy and Allergy and Dermatitis (PreventADALL) trial, showed similar results.²¹ A meta-analysis of these studies confirms the beneficial effect of early introduction of peanut in the prevention of peanut allergy with a high level of evidence.²⁷

Limitations in Implementation of Primary and Secondary Prevention

Introducing peanut to infants presents substantial challenges. Social beliefs, racial and socioeconomic factors, and limited awareness among health care professionals and parents create barriers to early introduction. Further obstacles to implementation include the nutritional effect of these interventions and the recommendation that infants be exclusively breast-fed for the first 6 months of life.

Health care professionals and parents need reassurance that it is safe to introduce the described quantities of peanut earlier and that introducing peanut will not affect continued breast-feeding or cause adverse nutritional outcomes.²⁸ The LEAP trial showed that even among infants consuming the highest quartile of peanut protein (median, 7.5 g per week), there were no differences in anthropometric measurements

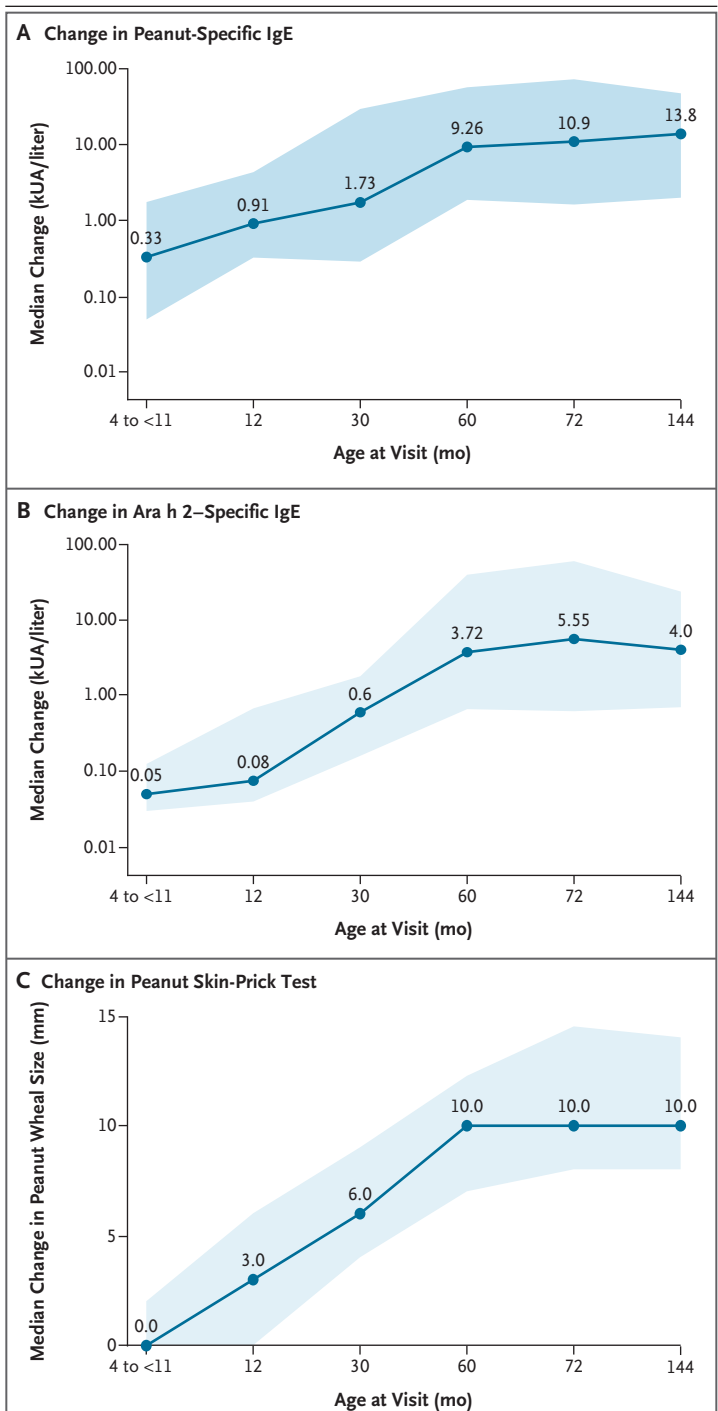


Figure 1. Change in Median Biomarker Levels in the LEAP, LEAP-On, and LEAP-Trio Cohorts.

Shown are changes in biomarker levels in the trial cohorts over the duration of the Learning Early About Peanut Allergy (LEAP) trial and two extensions of the LEAP trial: the Persistence of Oral Tolerance to Peanut (LEAP-On) trial and the Follow-up of LEAP Participants and Their Families (LEAP-Trio) trial.^{7,11,24} Shading represents interquartile ranges.

or energy intakes as compared with infants who did not consume peanut.²⁸

TERTIARY PREVENTION OF PEANUT ALLERGY

Peanut Oral Immunotherapy

Peanut oral immunotherapy is the oral administration of increasing doses of peanut protein in patients with peanut allergy. Randomized, controlled trials have shown peanut oral immunotherapy to be effective at increasing the dose threshold at which an allergic reaction occurs and reducing the reaction severity when it does occur.^{9,29,30}

In the Peanut Allergy Oral Immunotherapy Study of AR101 for Desensitization (PALISADE), 551 patients 4 to 55 years of age who had peanut allergy were randomly assigned to receive 300 mg of daily peanut protein or placebo. Among the majority of participants who were 4 to 17 years of age, 67.2% in the peanut-protein group were able to ingest a single dose of at least 600 mg of peanut protein in the double-blind, placebo-controlled food challenge conducted when they exited the trial with no more than mild symptoms, as compared with 4.0% in the placebo group ($P<0.001$).³¹

Peanut oral immunotherapy in younger children also appears to be safe and effective. The

Peanut Oral Immunotherapy Study of Efficacy, Immunologic Outcomes, and Desensitization (POSEIDON) investigated oral immunotherapy with daily oral peanut protein in 146 children with peanut allergy who were 1 to less than 4 years of age. After 24 weeks, 73.5% of the participants had clinical tolerance of a single dose of at least 600 mg of peanut protein with no more than mild symptoms in the double-blind, placebo-controlled food challenge administered on exit from the trial, as compared with 6.3% in the placebo group.⁹

The Induction of Tolerance and Mediation of Peanut Allergy in Children with Therapy (IMPACT) trial, which involved 144 children 1 to 3 years of age who had peanut allergy, showed an inverse relationship between the age of commencement of oral immunotherapy and clinical remission after cessation of treatment.³² Specifically, 71% of children who commenced treatment at less than 24 months of age were able to consume 5000 mg of peanut protein 6 months after cessation of treatment, whereas only 35% of those who were older than 24 months of age and 19% of those who were older than 36 months of age had the same level of remission 6 months after cessation.³² Children in whom tolerance was maintained had lower baseline levels of peanut-specific IgE and

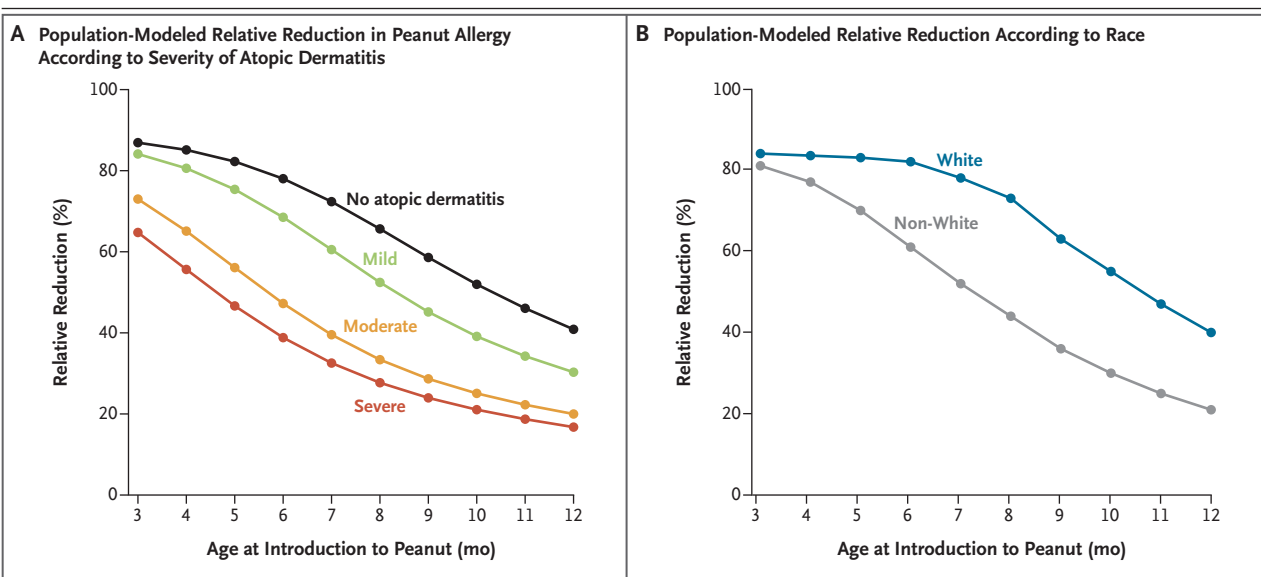


Figure 2. Modeled Effect of Delayed Peanut Introduction on the Development of Peanut Allergy.

Shown is the relative reduction in the burden of peanut allergy normalized to the age at introduction in a population stratified according to atopic-dermatitis severity (Panel A) and race (Panel B). The relative reductions estimate the treatment effect of early peanut introduction and avoidance. Figure modified and adapted (under a Creative Commons Attribution license) from Roberts et al.⁶

Table 1. Data Regarding Peanut Allergy from Major Trials on the Prevention of Food Allergy.*

Trial	No. of Participants	Atopic Dermatitis	Intervention	Race or Ethnic Group	Outcome Measures	Results for Avoidance vs. Consumption
LEAP ⁷						
Overall	640 infants at high risk	Severe	Early peanut introduction vs. avoidance Goal of 6 g peanut protein per wk	467 White (73%) 77 South Asian (12%) 64 Black (10%) 32 Other (5%)	Peanut allergy prevalence at age 5 yr	Allergy prevalence: 15.2% vs. 1.9% (difference, 87.5%) Allergy prevalence: 1.7% vs. 3% (difference, 81%)
Negative skin-prick test at baseline	542					13.7% vs. 1.9% (difference, 86%) 12.0% vs. 0.4% (difference, 96.6%)
Positive skin-prick test at baseline	98					35.3% vs. 10.6% (difference, 70%) 8.1% vs. 6.8% (difference, 75.8%)
EAT ²⁰						
	1303, general population	Included infants with or without atopic dermatitis	Early introduction of 6 allergenic foods: milk, peanut, sesame, fish, egg, wheat Treatment group: goal of 4 g of peanut protein per wk Control group: avoided peanut	1029 White (79%) 104 South Asian (8%) 91 Black (7%) 78 Other (6%)	Food allergy prevalence at age 3 yr	2.5% vs. 1.2% (P=0.11) 2.5% vs. 0% (P=0.003)
PreventADALL ²¹						
	2397, general population	Included infants with mild-to-moderate atopic dermatitis	Food-intervention groups: 1 lick of peanut butter 4–7 times per wk* Control groups: no consumption of peanut butter*	2037 White (85%) 144 South Asian (6%) 120 Black (5%) 96 Other (4%)	Allergy prevalence at 36 mo, severity of atopic dermatitis	2.0% vs. 0.9% 2.0% vs. 0.7%

* The EAT (Enquiring About Tolerance) trial and PreventADALL (Preventing Atopic Dermatitis and Allergies in Children) trial were multifood prevention trials; shown are the results with regard to peanut allergy from each. PreventADALL was a two-by-two intervention trial, but the results from the skin intervention did not show a treatment effect and are not shown. LEAP denotes Learning Early About Peanut Allergy.

Ara h 2-specific IgE and were younger (12 to 24 months of age) at commencement of oral immunotherapy.

Peanut oral immunotherapy can cause allergic symptoms. A 2019 systematic review showed that peanut oral immunotherapy was associated with increased risk and frequency of anaphylaxis, increased epinephrine use, and increased risk of serious adverse events and nonanaphylactic reactions as compared with placebo or avoidance.³³ However, the studies in this systematic review mainly included children who were older than 5 years of age. More recent studies have shown that there were less frequent and less severe adverse effects seen in younger children who received peanut oral immunotherapy.^{29,32} In the IMPACT trial, 98% of the participants who received peanut oral immunotherapy had at least one dose-related reaction, but no dose-related serious adverse events occurred; most reactions were mild (97%) or moderate (42%), with severe symptoms observed in only 5% of the participants.³² Similarly, no treatment-related severe adverse events were observed in POSEIDON, with 85% of the treatment-related adverse events classified as mild and 15% as moderate.⁹ Peanut oral immunotherapy has been associated with eosinophilic esophagitis in an estimated 2 to 3% of patients who received it.³⁴ Peanut oral immunotherapy is a lifelong treatment.

Peanut Epicutaneous Immunotherapy

Peanut epicutaneous immunotherapy was designed to limit intravascular peanut-protein absorption and systemic reactions. It consists of microgram quantities of lyophilized peanut protein in an occlusive chamber (known as a peanut patch) applied to the skin, initially for 3 hours and building up to 24-hour application. The Peanut EPIT (Epicutaneous Immunotherapy) Efficacy and Safety Study (PEPITES) was a randomized, controlled trial involving children 4 to 11 years of age (median, 7 years) who had peanut allergy; 356 children were enrolled, with 238 receiving a patch containing 250 μ g of peanut protein and 118 receiving a placebo patch for 12 months. The percentage of patients with a response (defined according to a binary criterion based on the change in a patient's eliciting dose from baseline to month 12) was 35.3% with the peanut patch and 13.6% with placebo ($P<0.001$).³⁵

The Epicutaneous Immunotherapy in Toddlers

with Peanut Allergy (EPITOPE) trial, which enrolled 362 children 1 to 3 years of age (median, 2.5 years) who had peanut allergy, showed that at 12 months after randomization, 67% of the participants in the peanut-patch group had protocol-defined desensitization, as compared with 34% of those in the placebo group ($P<0.001$).³⁶ In the open-label extension of the EPITOPE trial, 277 participants were reassessed for desensitization after an additional 12 months of treatment with a peanut patch. After receiving a total of 24 months of peanut epicutaneous immunotherapy, 84% of the participants met the criteria for a response, with 81% having tolerance when exposed to more than 1000 mg of peanut protein and 56% able to complete a 3444-mg double-blind, placebo-controlled food challenge without symptoms.³⁷

Trials of peanut epicutaneous immunotherapy showed few systemic side effects but a high incidence of local side effects at the application sites. Similar to peanut oral immunotherapy, epicutaneous immunotherapy had greater efficacy among younger children than older children.^{36,38} Epicutaneous immunotherapy is not approved by the Food and Drug Administration (FDA) for the treatment of peanut allergy.

Sublingual Immunotherapy

Sublingual immunotherapy involves the administration, under the tongue, of an allergen at gradually increasing doses that are lower than the doses used in oral immunotherapy. In a recent study involving 50 children (median age, 2.4 years), peanut sublingual immunotherapy was administered to 25 children and placebo to 25; after 36 months of receiving peanut sublingual immunotherapy at a maintenance dose of 4 mg of peanut protein, 79% of the participants in the active-treatment group were desensitized, as compared with 0% in the placebo group.³⁹ Desensitization was defined as a significant difference in the clinically tolerated dose of peanut during an oral food challenge at the end of treatment. Three months after treatment discontinuation, 63% of the original participants in the active-treatment group still had clinical tolerance, as compared with 0% of those who received placebo. Among younger children (1 to 2 years of age) and those with lower peanut-specific IgE levels at the time of enrollment, desensitization was observed in a higher percentage, and a higher

percentage had remission (78% at 1 to 2 years of age vs. 50% at >2 years of age).³⁹ Sublingual immunotherapy is not approved by the FDA for the treatment of peanut allergy.

Biologic Drugs

Biologic drugs are designed to block or modify specific immune system pathways involved in allergic responses. On the basis of findings in the Omalizumab as Utilization Monotherapy and as Adjunct Therapy in Children and High-Risk adults with Food Allergy (OUTMATCH) trial, omalizumab — which reduces free circulating IgE — was approved by the FDA for use in the treatment of persons 1 year of age or older who have multiple food allergies. The results of this trial showed that participants had a significant increase in tolerance to food allergens; specifically, 67% of participants treated with omalizumab were able to tolerate 600 mg or more of peanut protein (approximately 2.5 peanuts) as compared with 7% of those who received placebo ($P<0.001$).⁴⁰ As a passive immunotherapy, omalizumab does not modulate peanut-specific immune responses. Unlike oral immunotherapy, omalizumab is not allergen-specific and does not require strict avoidance of exercise or exposure to hot water for 2 hours after administration of the peanut dose. However, long-term continuous treatment is recommended, because there is no evidence that protection continues after withdrawal of treatment. Other biologic agents and small molecules — such as Bruton's tyrosine kinase inhibitors, some of which are licensed for use in the treatment of other allergic diseases — are under investigation for use in persons with food allergy.

AREAS OF UNCERTAINTY

Although meta-analyses of randomized, clinical trials of early consumption of peanut and egg in infancy show high levels of evidence for prevention,⁴¹ and findings in observational cohort studies suggest that early consumption of dairy may prevent allergy to cow's milk,⁴² studies involving other food allergens are lacking. The precise quantity and duration of peanut protein required to induce and maintain oral tolerance is not yet established. It is probable that the appropriate amount of consumption in infants varies on the basis of their baseline risk profile; currently, 2 g

weekly is suggested for infants at low risk and 4 to 6 g for those at higher risk. There is a concern that early consumption of microgram-to-milligram quantities of allergen protein may promote allergy rather than tolerance, particularly if exposure is infrequent.

A profound disparity exists in the prevalence of peanut allergy among non-White infants in high-income nations — 29.2% among Asian infants, 14.6% among Black infants, and 14.7% among infants of mixed race, as compared with 5.5% among White infants.⁶ Peanut allergy tends to develop earlier in the first year of life in non-White infants than in White infants.⁴³ The underlying drivers of this increased susceptibility — whether genetic, environmental, or related to migration — remain unclear, which indicates an urgent need for targeted research and support for these families.

The role of skin-barrier repair in preventing transcutaneous sensitization is under investigation. The PACI (Prevention of Allergy via Cutaneous Intervention) trial showed that intensive intervention (emollients plus topical glucocorticoids) within the first 12 weeks of life yielded a 25% relative reduction in egg allergy at 1 year of age.⁴⁴ However, this outcome was accompanied by significant decreases in weight gain and linear growth. Conversely, studies of the use of emollients alone have failed to show prevention of food allergy. The identification of safer, more effective strategies to maintain skin-barrier integrity and prevent the development of atopic dermatitis is a research area of increasing interest.

Currently, peanut allergy cannot be cured, as shown by studies of oral immunotherapy and sublingual immunotherapy in which, after a prolonged treatment period, clinical remission can be observed for up to 6 months after treatment cessation, but by 1 year after treatment cessation the majority of the treatment population is no longer protected.^{32,39} Curing peanut allergy remains a challenge. Two primary avenues are currently being explored: early intervention by initiating immunotherapy before 12 months of age, and combination therapy with biologics and other molecules. The COMBINE (Combination Omalizumab and Multi-Allergen Oral Immunotherapy Biologics to Improve New Efficacy) trial (ClinicalTrials.gov number, NCT03679676), which is exploring the use of omalizumab and dupilumab as adjuncts to multifoed oral immuno-

KEY POINTS

PREVENTION AND TREATMENT OF PEANUT ALLERGY

- Peanuts should be introduced at 4 to 6 months of age in infants at high risk for allergy and at approximately 6 months of age in other infants.
- The window of opportunity for prevention is narrow, and every month of delayed introduction increases risk, particularly for infants who have severe atopic dermatitis or who are of non-White race.
- Appropriate prevention involves the ingestion of approximately 2 g of peanut protein weekly for infants at low risk, whereas infants at high risk should receive higher doses (4 to 6 g per week).
- Peanut immunotherapy is more effective, safer, and more likely to induce clinical remission in children in whom it was initiated at an early age than in children who began immunotherapy later.
- The natural history of peanut allergy follows a trajectory of increased allergic immune responses to peanut and increased clinical reactivity over time.
- Public health interventions that involve the early introduction of peanuts for all infants are likely to prevent peanut allergy in most infants and to enable earlier diagnosis and treatment in those who have peanut allergy; this approach is expected to lead to earlier and better treatment outcomes.

therapy, is an example of ongoing efforts to determine whether these combinations can induce long-term, treatment-independent tolerance.

GUIDELINES

Previous guidelines for the prevention of allergy have recommended that infants should not be exposed to peanuts before specific ages (in the first year of life in the United Kingdom and not until 3 years of age in the United States). Current recommendations by national and professional societies in the United States²² and by the European Academy of Allergy and Clinical Immunology (EAACI)²³ state that peanuts should be introduced at 4 to 6 months of age in at-risk infants and at approximately 6 months of age in other infants. The 2021 North American Consensus guidelines recommend early introduction of peanut, egg, and other food allergens during early weaning to prevent food allergies.⁴⁵ This approach is also reflected in the recent Department of Agriculture and Department of Health and Human Services Dietary Guidelines for Americans.⁴⁶ The introduction of peanut and other food allergens at 4 to 6 months of age conflicts with the World Health Organization (WHO) recommendations for exclusive breast-feeding until 6 months of age,⁴⁷ which aim to maximize nutrition and reduce infectious-disease burden in low-income countries (where rates of food allergy are generally very low). Strict adherence to WHO guidelines may be contributing to the continued rise in food allergies in high-income countries. Recommendations in this article are generally

consistent with national and professional society guidelines.

Guidelines from the EAACI recommend oral immunotherapy as a therapeutic option for children and adolescents with peanut allergy and suggest oral immunotherapy for peanut, milk, and egg allergies.⁵ Omalizumab is recommended for use in the United States and by the EAACI to treat adults and children 1 year of age or older who have confirmed IgE-mediated food allergy.^{5,48}

CONCLUSIONS AND RECOMMENDATIONS

The infant described in the case vignette was of East Asian heritage but was born and living in the United States, had early-onset severe atopic dermatitis, and had consumed peanuts for the first time at 7 months of age — an age that is too late in an infant who is at such a high risk for peanut allergy. Earlier introduction of peanut — at 3 to 4 months of age — and frequent consumption would have been necessary to affect the development of peanut allergy. Even in children at lower risk for peanut allergy, a delay in the introduction of peanut reduces the efficacy of this prevention strategy, and early consumption should be encouraged in all infants.

In the past, the default management strategy for an infant with peanut allergy was to advise strict avoidance of peanuts and constant availability of emergency medications, typically including an epinephrine autoinjector, as part of a bespoke emergency plan. Given that immunotherapy is more likely to be effective and result

in greater tolerance at an early age, we would recommend oral immunotherapy for this infant. We would also encourage the frequent ingestion of other common food allergens relevant to the family's diet to ensure allergy prevention and ongoing tolerance. Although current therapies have a positive effect on quality of life for persons with peanut allergy, there is as yet no cure that permits an end to ongoing treatment.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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